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CHAPTER - 1

INTRODUCTION

To offer people with extreme disabilities, an opportunity to control a computer simply by moving his/her eyes or head.To design a low-cost combined eye and head tracking system for persons with deficiency of their upper limbs.

“Eye mouse” is similar to conventional computer mouse, but entirely controlled by the eyes.

A real time "gaze" determination software that controls a computer cursor by following the user's gaze.

A while back I mentioned that I was working on a an Eye-Tracking Project called Pupil with a couple of friends of mine. It’s come a long way now and we’ve have had considerable success - but that’s for another post. Pupil is a great tool for eye-tracking and getting gaze data that can be used with other applications in real time. It utilizes two HD cameras mounted on an eye-glasses frame at 180˚ from each other. One is called ‘Eye Camera’ that records your eye movements and the other is called ‘World Camera’ which is used to process exactly where the user is looking at. This data can be streamed to other applications over TCP in real time.

We wanted to control the mouse using our eye gestures, but it heavily relied on the World Camera and defining virtual screens called surfaces within the application. This obviously resulted in greater accuracy since it moved the pointer only when you actually looked at the screen.

CHAPTER - 2

DATA FLOW DIAGRAM

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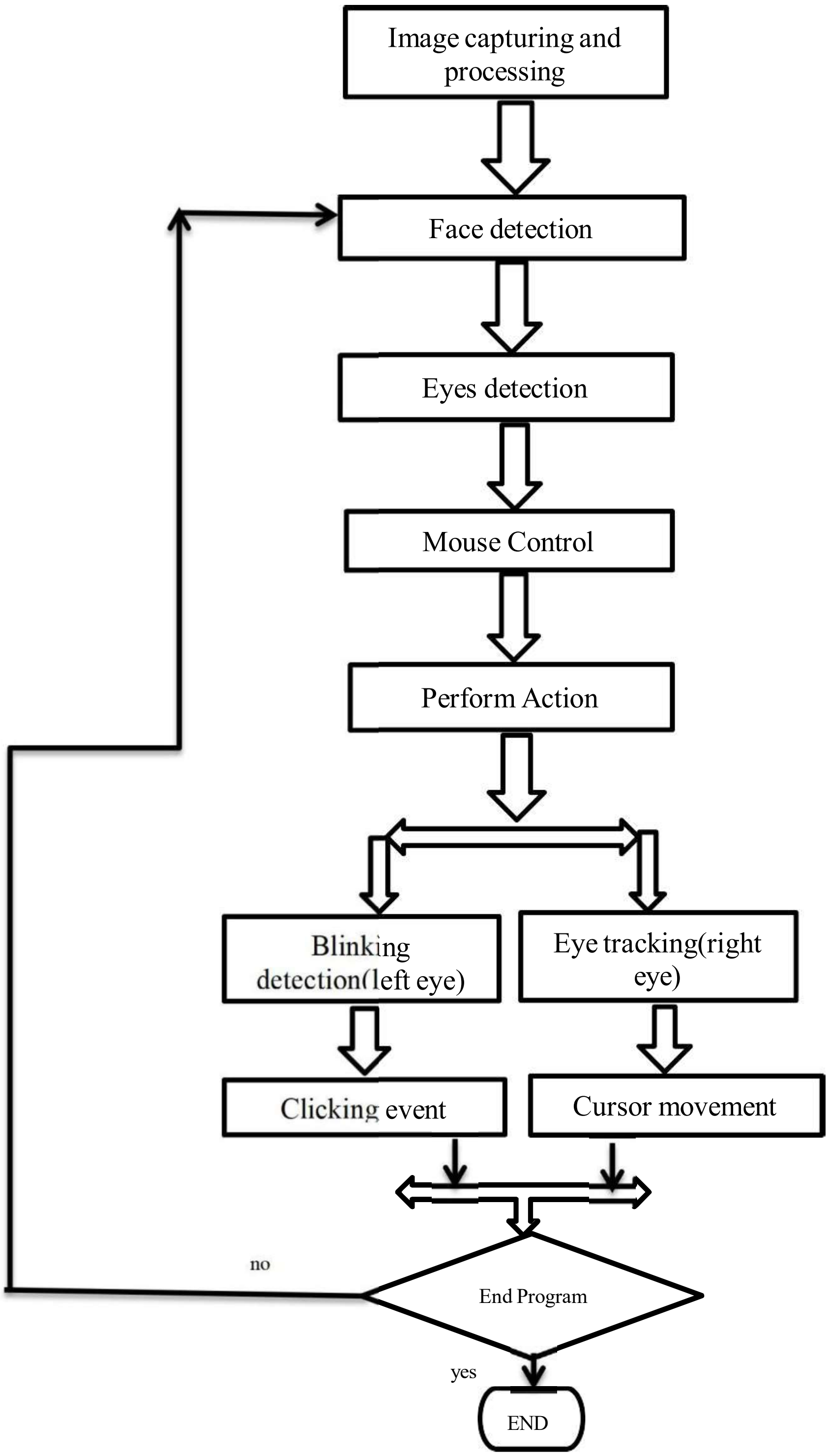

CHAPTER - 3

3.1) Description of PYTHON

- Python is a simple, general purpose, high level, and object-oriented programming language.
- Python is an interpreted scripting language also. *Guido Van Rossum* is known as the founder of Python programming .
- It supports object-oriented programming, procedural programming approaches and provides dynamic memory allocation.
- Python has wide range of libraries and frameworks widely used in various fields such as machine learning, artificial intelligence, web applications, etc.
- Python can be a powerful tool in developing these NLP skills and understanding how people search and how search engines return results.
- Python is even used in 3D animation software such as Lightwave, Blender, and Cinema 4D, showing just how versatile the language is.

CHAPTER - 4

OUTPUT SCREENS

Figure 1 : Image capturing and Processing

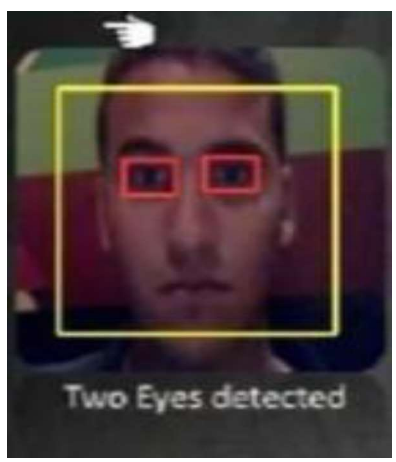

Figure 2 : Detects the both eyes

Figure 3 : When both the eyes are not detected then webcam will off .

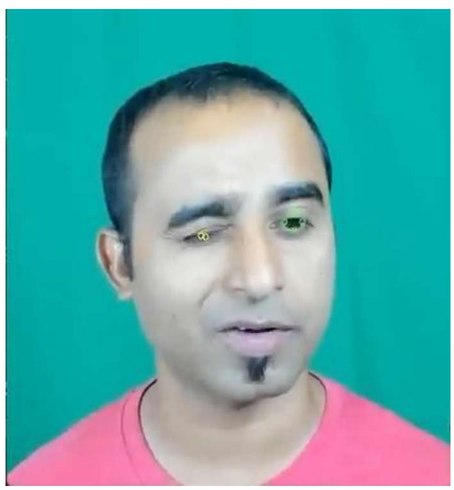

Figure 4 : Blinking detection

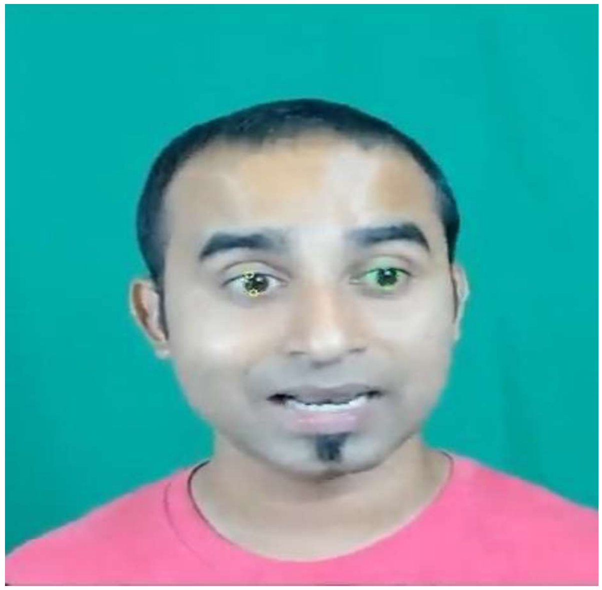

Figure 5 : Eye tracking

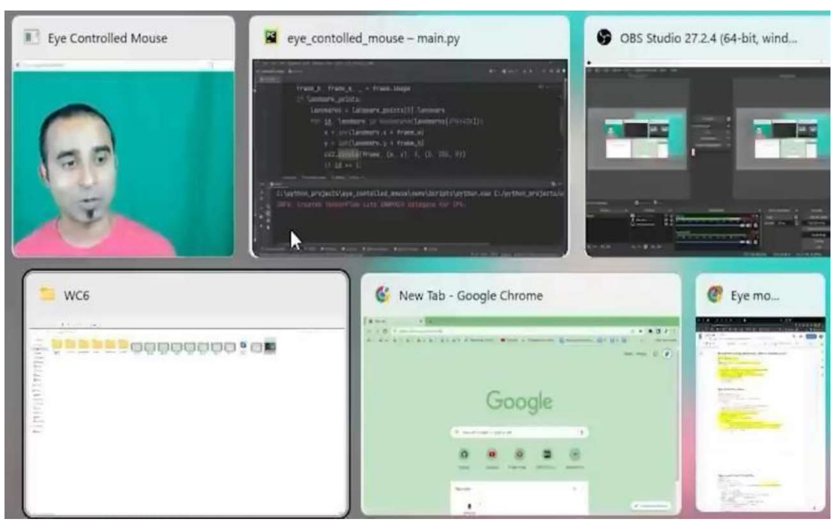

Figure 6 :When we have go to the next tab with the help of eye controlling mouse

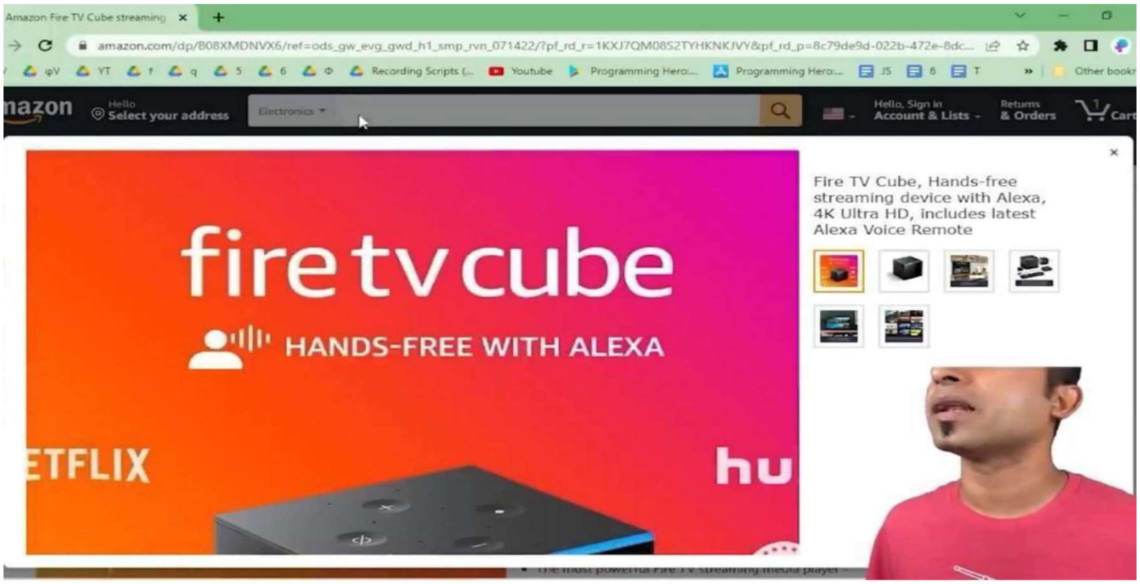

Figure 7 : When we have to search in the search box of the website.

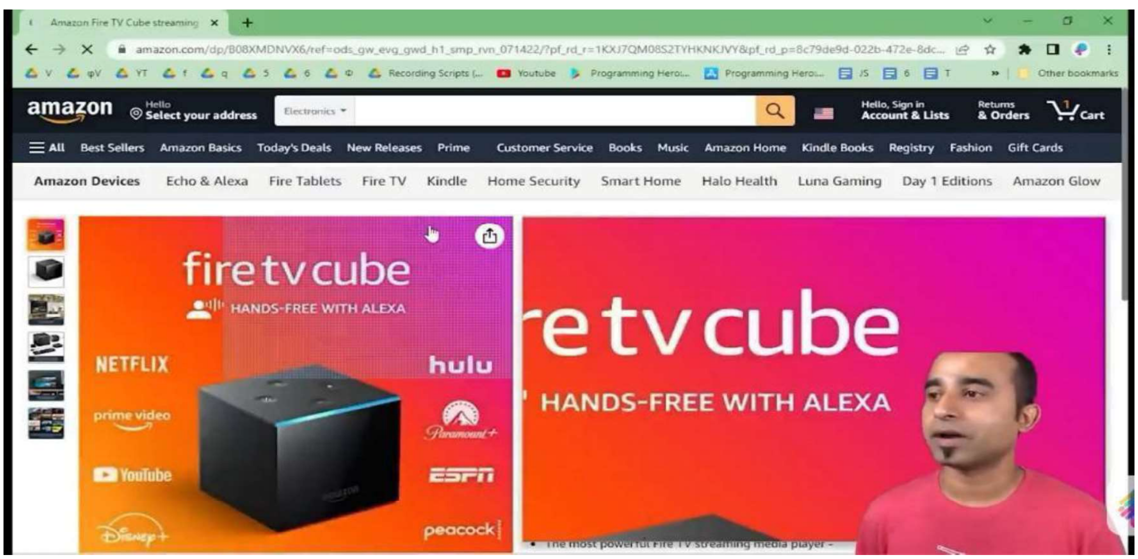

Figure 8 : When we have to go anywhere with the help of eye controlled mouse in the website.

IMPLEMENTATION CODE

#PYTHON CODE

```
import cv2

import mediapipe as mp

import pyautogui


cam = cv2.VideoCapture(0)

face_mesh = mp.solutions.face_mesh.FaceMesh(refine_landmarks=True)

screen_w, screen_h = pyautogui.size()


while True:

    _, frame = cam.read()

    frame = cv2.flip(frame, 1)


    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    output = face_mesh.process(rgb_frame)

    landmark_points = output.multi_face_landmarks

    frame_h, frame_w, _ = frame.shape
```

```
if landmark_points:
    landmarks = landmark_points[0].landmark
    for id, landmark in enumerate(landmarks[474:478]):
        x = int(landmark.x * frame_w)
        y = int(landmark.y * frame_h)
        cv2.circle(frame, (x, y), 3, (0, 255, 0))
        if id == 1:
            screen_x = screen_w * landmark.x
            screen_y = screen_h * landmark.y
            pyautogui.moveTo(screen_x, screen_y)
        left = [landmarks[145], landmarks[159]]
        for landmark in left:
            x = int(landmark.x * frame_w)
            y = int(landmark.y * frame_h)
            cv2.circle(frame, (x, y), 3, (0, 255, 255))

        if (left[0].y - left[1].y) < 0.004:
```

```
        pyautogui.click()
        pyautogui.sleep(1)

cv2.imshow('Eye Controlled Mouse', frame)
cv2.waitKey(1)
```

REFERENCES

1. W3 Schools
   https://www.w3schools.com

2. Python official website
   https://www.python.org

3. Javatpoint
   https://www.javatpoint.com

4. GeeksforGeeks
   https://www.geeksforgeeks.org

5. Mozilla developer
   https://developer.mozilla.org